

(CSA0711) 29

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I, Pranav / 190317042 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pranav

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

24/10 Pranav
Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

CO2 - BT - 3

II Debugging + optimisation Task:-

Given configuration:-

IP addresses : 192.168.10.65

Subnet mask : 255.255.255.192 (/26)

Default Gateway : 192.168.10.1

Required subnet : 192.168.10.0/26

i) Identify the subnet to which the configured IP address belongs.

192.168.10.0 - 192.168.10.63

192.168.10.64 - 192.168.10.127

192.168.10.128 - 192.168.10.191

192.168.10.192 - 192.168.10.255

ii) Network address, Broadcast address, and valid Host Range.

Parameter	Value
Network address	192.168.10.64
Broadcast address	192.168.10.127
Valid Host Range	192.168.10.65 - 192.168.10.126

iii) Verify whether the configured IP address and Gateway are valid

IP address

Configured IP = 192.168.10.65

Required subnet = 192.168.10.0/26

Known:-
 192.168.10.45 belongs to 192.168.10.0/24, net 192.168.10.0

(iv) Configuration Error:-

Treated IP address

192.168.10.45 belongs to a different subnet

IP address and gateway are in different subnet

Host: 192.168.10.45/24

Gateway: 192.168.10.0/24

v) optimized configuration:-

Parameters	correct value
IP address	192.168.10.2
Subnet mask	255.255.255.192 (/26)
default gateway	192.168.10.1

vij) Total number of usable Host addresses after conversion
 $2^{(32-26)} - 2 \Rightarrow 2^6 - 2 \Rightarrow 64 - 2 = 62$

2. Debugging & optimization:-

Problem:-

Device A:- 2001::db8::ff00:42:8329/64

Device B:- 2001::db8::0:0:1::1/64

1. Expand both IPv6 addresses:-

Device A:-

Expanded:- 2001::db8::ff00:42:8329

Expanded:- 2001::db8::0000:0000:0000:ff00:0042:8329

B:-

Expanded:- 2001::db8::0:0:1::1

Expanded:- 2001::db8::0000:0000:0001:0000:0000:0001

Summarize the Network prefix:-

1) a /64 prefix the first 64 bits (first four hexadecimal blocks) must be the network prefix.

2) Network prefix

Device A: 2001::db8::0000:0000:1::1/64

Device B: 2001::db8::0000:0000:1::1/64

Verify whether both devices belong to the same subnet

Both devices have the same /64 network prefix

2001::db8::0000:0000:1::1/64

Yes, both devices belong to the same IPv6 subnet.

3) optimize the IPv6 configuration:-

Assign unique addresses within the same /64 subnet

Device optimized IPv6 address

Device A: 2001::db8::0:0:1::10/64

Device B: 2001::db8::0:0:1::20/64

Both devices now use the same network prefix

2001::db8::0:0:1::10/64

connected to the ISP or Internet.

company :-

①-

Administration LAN

Network : 192.168.10.0/26

Gateway : 192.168.10.1

Finance LAN

Network : 192.168.10.64/26

Gateway : 192.168.10.65

Subnet mask for /26 : 255.255.255.192

Each /26 Subnet contains

64 IP addresses

62 usable host addresses.

i). determine the Network address and Broadcast address.

LAN	Network address	Broadcast address
Administration	192.168.10.0	192.168.10.63
Finance	192.168.10.64	192.168.10.127

ii) Valid Host IP address Range.

LAN	Valid Host Range.
Administration	192.168.10.1 - 192.168.10.62
Finance	192.168.10.65 - 192.168.10.126

iii) Suggested IP addresses for three computers.

Computer	IP address
PC-1	192.168.10.2
PC-2	192.168.10.3
PC-3	192.168.10.4

Gateway 192.168.10.1

Finance LAN

Computer	IP address
PC-1	192.168.10.66
PC-2	192.168.10.67
PC-3	192.168.10.68

Gateway : 192.168.10.65

4. Default Gateway for Each LAN :-

LAN	Default Gateway
Administration	192.168.10.1
Finance	192.168.10.65

The default gateway is the router interface that enables communication with other networks.

vi) calculate the no. of Available Host addresses

IPv6 addresses length = 128 bits

Prefix length = 64 bits

Host bits :- $128 - 64 = 64$

$2^{64} = 18446744073709551616$

3. Debugging & Optimization Task - VLSM Subnet Allocation

Given:-

Network : 192.168.1.0/24

Subnet masks used :- /26, /27, /28

i) Network address, Broadcast address, and host range.

Subnet	Network address	Host range
192.168.1.0/26	192.168.1.0	192.168.1.1 - 192.168.1.62
192.168.1.64/27	192.168.1.64	192.168.1.65 - 192.168.1.94
192.168.1.96/28	192.168.1.96	192.168.1.97 - 192.168.1.110

Broadcast - 192.168.1.63, 192.168.1.95, 192.168.1.111

usable hosts - 62, 30, 14

ii) no. of usable and unused IP addresses

Department A : B : C :- 50 : 20 : 10

A	/26	62	50	12
B	/27	30	20	10
C	/28	14	10	4
Allocated subnet		usable Host	Hosts Required	unused Hosts

Identify the network prefix using the /64 prefix length:-

A/64 prefix means:-

First 64 bits - network prefix

Last 64 bits - Interface Identification (Host ID)

Network prefix

2001:0db8:abcd:0::/64

iv) calculate the total number of Host addresses supported within the subnet

Since the prefix length is /64:-

$$\text{Host bits} = 128 - 64 = 64 \text{ bits}$$

$$\text{Total host addresses:- } 2^{64} = 18446744073709551616$$

v) Tot no. of possible Host addresses Available within this network

IPv6 does not reserve separate network and broadcast addresses as IPv4 does. Therefore, all 2^{64} addresses are usable.

4. An Internet service provider (ISP):-

Given:-

Routing table

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Subnet Mask

255.255.255.0

255.255.255.0

255

Destination IP address:- 192.168.2.45

i). determine the destination subnet:-

A/24 subnet means the first 24 bits (first three octets)

Destination IP address:- 192.168.2.45

Network portion:- 192.168.2.0/24

Answer:- destination subnet: 192.168.2.0/24

ii) Matching Routing Table Entry:-

compare the destination network with the routing table

Routing Entry	Match
192.168.1.0/24	-
192.168.2.0/24	✓
192.168.3.0/24	-

iii) Next-Hop Network / interface:-

outgoing Network: 192.168.2.0/24

Next Hop Interface: Interface connected to the 192.168.2.0/24

iv) Network (e.g., Gigabit Ethernet 0/1, depending).

v) Default Route:-

if the destination IP does not match any routing table entry, the router uses the default route.

Default route:- 0.0.0.0/0

The packet is forwarded to the default gateway (next-hop)